7.1 Interface Layout and Controls

- Search by callsign or ICAO address and select aircraft from the map or traffic list.
- Use map zoom and center controls; inspect flight, aircraft and route tabs in the lower panel.
- Read live/simulation mode, connection health, prediction age and altitude history without leaving the dashboard.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project implements a complete local pipeline from PingStation receiver or replay input to decoded aircraft records, persistent trajectory history and browser visualization. The tested source compiles successfully, the dashboard JavaScript passes syntax checking, and short receiver gaps are bridged by clearly labelled regression or kinematic predictions. The system is appropriate for education, software experimentation and trajectory-data study; it is not certified surveillance equipment.

8.2 Future Scope

- Evaluate longer live-receiver sessions and report quantitative error metrics against trusted reference tracks.
- Compare the current linear baseline with Kalman filtering [10], tree ensembles [11] and recurrent models [12]-[14].
- Add quality dashboards, altitude/time filters, authentication and controlled network deployment.
- Investigate route- and weather-aware trajectory models using a larger verified dataset.

APPENDICES

- Appendix A: Configuration Summary
- Appendix B: Verified Main Code Listings
- Appendix C: Project File Inventory
- Appendix D: Sample Output Records
- Appendix E: Execution Steps
- Appendix F: Sample Raw PingStation Log
- Appendix G: Algorithm Selection and Comparison
- Appendix H: User Interface Screenshots